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NEET-UG 2026 was cancelled nine days after 22.79 lakh candidates sat it.

3 May → 12 May

held, then cancelled after a leak

~45 hours

head start the leaked "guess paper" had

CBT from 2027

NEET moves to computer-based testing

Every failure becomes a nationwide re-test

- NEET-UG 2024: 1,563 candidates got grace marks; they were scrapped, a re-test offered, and only 813 took it.
- The Supreme Court called the 2024 leak localised (~155 beneficiaries), but had no fine-grained evidence to separate them.
- Wrong paper sets sent to centres, papers moved by e-rickshaw, weak CCTV (recorded by the Court).
- NEET needs ~22 lakh seats vs 2–2.5 lakh CBT seats per session: multi-shift, so fairness evidence is core.

813 / 1,563

took the 2024 re-test they were offered

5 months

to investigate + try under the 2026
Amendment Bill

Not "zero failures". Every failure is:

1

Recoverable

No acknowledged answer is ever lost, seat, relay and data centre can each die.

2

Contained

Blast radius is shown live; only the harmed candidates are re-tested.

3

Provable

Tamper-evident records every candidate can verify offline.

Five parts, each distrusting the next



- Answers are WhatsApp-style ticks: ✓ relay has it, ✓✓ data centre has it, blue = countersigned.
- Paper keys are a two-key locker: any 2 of 3 custodians (NTA, NIC, observer).

Witness
co-signs the log (built;
demo runs it beside
control)

Risky seats caught before T0

- Seat A running AnyDesk and an overlay tool → gate names both and blocks.
- Close them, press Re-check → green; all three seats start.
- Scribe seat: 2 faces expected, 3 faces raise one flag that a human reviewer clears.
- In-exam AnyDesk → an integrity entry, never an auto-submit; the seat keeps answering.

2 of 2

tools named and blocked (AnyDesk, overlay-sim)

5 of 5 → 1 flag

3-face samples → one face-extra flag, human-cleared

The exam starts on time, even offline

- We degrade Centre 42's link: SYNC_LAG warns before the cut.
- Two of three custodians approve; one alone cannot release the paper.
- The superintendent types a phoned code: it unlocks only Centre 42, this shift. A typo or another centre's code fails.
- Before T0 the paper is ciphertext on disk; a forged key from the relay is rejected.

1.0–1.9 s

SYNC_LAG warning before the WAN cut (3 runs)

2 of 3

custodians needed to release paper keys

We pulled the plug on a data centre

0

answers lost, P1 closed with lost 0

6,593

entries: sent = verified = stored

1.3–1.9 s

recovery time (RTO), RPO 0, 3 chaos runs

- Cell 2 killed with SIGKILL and its database deleted; it rebuilt from the relays.
- Candidates kept answering; the blast radius (1 cell · 2 centres · 100 candidates) was shown live.
- A spare relay took over with nothing lost.

An insider edits an answer. We catch it.

Q17: record says C, the seat committed B

- Receipt code computed on the seat at submit, like a UPI reference.
- /verify is one offline 84 KB file: no network, no keys.
- One-click evidence pack with a BSA s.63 certificate template.

1,000 / 1,000

one-byte flips located at their entry

30 / 30

golden-vector self-checks on every /verify load

Only the harmed are re-tested

Re-tested 225 • Re-conducted 3 centres • Spared 18,994 • ₹ avoided 2,84,91,000

19,811

live candidates, 20k run

1,981,100

export rows analysed

322

flags, each explained

0.00%

honest false-positive rate
(G2)

- Radar runs on what actually flowed through the cells; the decision applies the NEET-2024 Supreme Court tier-1 test.
- A named human signs the report hash. The signature records approval, it does not make the report correct.

History corroborates. It never accuses.

Flags come from the exam

Every flag shows its reason, observed vs expected, and a p-value.

History only annotates

Past performance annotates flags already queued, never creates or escalates one (322 annotated, 55 corroborated).

Look-alikes protected

Hindi-medium, PwD, rapid guessers and improvers are not flagged.

Nothing auto-penalised

A human decides; time is credited by the relay clock, capped at 30 min.

Limits and roadmap

- Deterrence plus detection, not cheat-proof; records are tamper-evident.
- Integrity gate detects by process name; the camera face check is run by hand.
- Radar and risk model are proven on synthetic cohorts; field accuracy comes from a pilot.
- Next: field pilot on a real WAN, power-loss and low-end seat tests (20k ran on one Mac, loopback).
- Next: HSM-held sign-off key and real WORM archive (today: DEV key, simulated WORM).
- Next: independent-custody witness and pinned TLS between components.

Measured numbers

Measure	Result	File (docs/evidence/)
Ingest latency, relay to cell	p50 44 ms • p99 168 ms	stage7-load.txt
Stored / rejected (20k cohort)	227,882 entries • 0 rejected	stage7-load.txt
Candidates unlocked	19,803 / 19,803	stage7-load.txt
Relay traffic per candidate-hour	≈1.06 MB (60 s window extrapolated)	stage7-load.txt
Seat CPU / RSS (Node main process)	2.3% median • 166 MiB	stage7-load.txt
Chaos drill, 20 runs	0 of 16,000 answers lost • RTO medians 0.96–1.23 s	stage8-chaos.txt

One Apple M5 Mac, 3 real cells on loopback, 99 simulated centres, synthetic cohort. Seat figures cover the Node main process only; not real WAN.

A protocol, not a product

- Vendor conformance kit in NTA's RFP: any vendor's seat, relay or cell must pass the same chaos and tamper tests.
- RACI: NTA owns policy, keys and decisions; NIC runs control and cells; vendors run centres; witnesses sit outside the operational path.
- Tested on macOS and Windows.
- Apache-2.0 under the GoI open-source policy.
- Rollout: shadow mode on a small NTA exam → relays and cells → mandate.

All 12 focus areas, traced to code and tests

1 Real-time monitoring	Partly	7 Reconciliation	Built
2 Early failure prediction	Partly	8 Candidate communication	Built
3 Incident detection & escalation	Built	9 Re-conduct decision	Built
4 Backup and DR	Partly	10 Fairness when disrupted	Partly
5 Tamper-evident storage	Built*	11 Audit trail & evidence	Built*
6 Suspicious patterns	Partly	12 AI systemic-risk analytics	Built

Source: docs/traceability.md summary table. * witness built; the demo runs it beside control. "Partly" rows list their remaining work there.